

# Pallavi Reddy Rangasamudram


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**Available:** From December 2026, full-time


## PROFESSIONAL SUMMARY


Test and validation engineer completing an M.Eng. in Electrical and Microsystems Engineering at OTH Regensburg, with almost two years at an automotive access-systems supplier across cybersecurity, development tooling and test engineering. Master thesis: a robotic test bench for door-handle ECUs, from arm assembly and sensor integration to closed-loop CAN testing in Vector CANoe and probe testing from $-40\,^{\circ}\mathrm{C}$ to $+80\,^{\circ}\mathrm{C}$. Earlier: business automation software at Persistent Systems, India.


## PROFESSIONAL EXPERIENCE

**Minebea Access Solutions Deutschland GmbH** — Erdweg, Germany

***Master Thesis, Test Engineering – Robotic Test Automation for Door Handle ECUs*** — Oct 2025 – May 2026

*Analysis and improvement of the Lock/Unlock test automation strategy with an ECU-controlled robotic arm*

- Identified a finger-simulation probe that matched human capacitive response across $-40\,^{\circ}\mathrm{C}$ to $+80\,^{\circ}\mathrm{C}$, by testing six probe configurations at 1–5 N contact force, grounded and ungrounded, against measured human-finger reference values for an OEM door handle.
- Automated the capacitive Approach and Unlock tests with up to two autonomous retries, confirmed in an R&R study, by building a closed-loop CAN setup: C/C++ firmware on an Arduino GIGA (ARM Cortex-M4, TJA1051 transceiver) and CAPL evaluation of ECU signals in Vector CANoe.
- Brought a 3D-printed 6-axis robotic arm from kit to repeatable target-position replay (five axes integrated, four verified), by assembling the drive train, wiring stepper drivers, magnetic encoders and limit switches, and resolving gear binding by reprinting at adjusted scale.
- Defined an improved test strategy for push, pull, Lock and Unlock actions, after analysing the limits of the existing 3-axis tester (LabVIEW runtime, NI DAQ, PCAN, CAN/LIN/K-Line).
- Integrated NFC key-card, capacitive and strain gauge (force) sensors into the test environment; flashed OEM handle ECUs for each software release and extended the existing CAPL test library for the feedback loop.
- Alongside the thesis: ran functional validation tests with the validation team on door handles after climatic chamber exposure and at $60\,^{\circ}\mathrm{C}$, recording error values per handle and action; completed in-house soldering training.

***Working Student – Electronics Development*** — Jan 2025 – Sep 2025

- Reduced manual planning work for project managers and team leads, by configuring and administering Codebeamer ALM and building tracker automation, a resource planning tracker for team availability, and time and leave planning templates.
- Took over Codebeamer administration and worked fully independently within a short time.

***Student Intern / Project Thesis – Automotive Cybersecurity*** — Jul 2024 – Oct 2024

- Carried one Cybersecurity Assurance Level (CAL) from stakeholder requirements through six ASPICE V-model layers to software detailed design, verified on a sample trace chain, by building CAL fields, reference links and UEL computed fields with a highest-CAL rule in Codebeamer (project thesis, grade 1.3).
- Maintained the Cybersecurity Case of an OEM project per ISO/SAE 21434 and UNECE R155: restricted TARA status changes, restored links to the standard and updated CSMS documentation and traceability.
- Supported the S/MIME secure e-mail rollout for OEM project members (certificates, private keys), reported issues to the vendor and handled certificate requests and revocations via GLPI tickets.

**Persistent Systems Ltd.** — Pune, India

***Associate Software Engineer (previously Student Intern)*** — Feb 2022 – Nov 2022

- Replaced manual registration and record-keeping steps for client business units, by developing business process automation applications and interfaces in Appian with SQL and Python.

**Primark**, Regensburg | *Sales Assistant (part-time), daily customer communication in German* — Jun 2026 – Present

## EDUCATION

**Ostbayerische Technische Hochschule (OTH) Regensburg** — Regensburg, Germany

*M.Eng. Electrical and Microsystems Engineering; preliminary GPA 2.5* — Oct 2023 – expected by Mar 2027

Project Thesis graded 1.3;
modules incl. Advanced Semiconductor Technology, Advanced Packaging, Advanced Optical Sensing

**Sreyas Institute of Engineering and Technology (JNTUH)** — Hyderabad, India

*B.Tech Electronics & Communication; GPA 2.2, First Class with Distinction* — Aug 2018 – Jul 2022

## PROJECTS AND SEMINARS

- **Smart Car Parking Control System** (B.Tech project): Arduino-based gate control with entry and exit sensors, a servo-driven barrier and an LCD showing free spaces.
- **Cyber Threat Detection with Artificial Neural Networks** (B.Tech project): neural-network classification of security events based on event profiles.
- **Seminars:** Thin Gate Oxide Layers in Semiconductor Devices (M.Eng.); Hadoop Distributed Storage and Processing (B.Tech).

## TECHNICAL SKILLS

- **Test & Validation (Prüfung, Validierung):** HIL test setups, Functional testing, Climatic and temperature testing, Test procedures, R&R studies, Test reports, Failure analysis
- **Vehicle Electronics:** CAN, Vector CANoe, CAPL, ECU flashing, NFC and capacitive access systems
- **Hands-on & Integration:** Robotic arm assembly and commissioning, Wiring, Sensor integration (capacitive, NFC, strain gauge, encoders), Stepper drivers, Soldering, 3D printing
- **Embedded & Programming:** C/C++, ARM Cortex-M4, Arduino, UART, Python (basic), SQL
- **Measurement:** Multimeter, Oscilloscope, Logic analyser
- **Tools & Processes:** Codebeamer ALM, Traceability, V-model, ASPICE, ISO/SAE 21434, UNECE R155, GLPI, Fusion 360 (model review), Microsoft Office
- **Familiar with:** LabVIEW / NI DAQ test systems, LIN, K-Line, ROS2 (self-study)

## CERTIFICATIONS AND WORKSHOPS

- Secure Hardware Components in Automotive, Strategische Partnerschaft Sensorik e.V., Regensburg 2024
- Appian Certified Associate Developer 2022
- Microsoft Certified: Azure Fundamentals (AZ-900) 2022
- Workshops: PCB Designing (2019), Internet of Things (2019), Projects using Cisco Packet Tracer (2021)

## LANGUAGES

**English** (C1, fluent); **German** (A2 completed, B1 course in progress, daily spoken use at work); **Telugu** (native)

## INTERESTS AND VOLUNTEERING

- **Regensburg Women's Cricket:** team player in regional tournaments; volunteer organiser of cricket promotion sessions at OTH Regensburg.
- **Campus Radio:** hosted and scripted English programmes during the B.Tech.
- **Community Service:** organised learning and nutrition-awareness activities for children.